Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia

Week 7 — Peer review: how it works

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Today

1. Feedback and Revision
2. Peer Review, Trust, and Metrics
3. Problems: Bias & Inequality, (un)Reproducibility

Roadmap

1. Feedback and Revision
2. Peer Review, Trust, and Metrics
3. Problems: Bias & Inequality, (un)Reproducibility

Why feedback?

- First drafts are rarely clear, complete, or well-argued
- Feedback (from peers, tutors, self-reflection) helps identify:
  - unclear claims
  - logical gaps
  - missing evidence
  - stylistic issues
- Better writing is clearer and more persuasive, and when you write about people and their data, more responsible

What feedback gives you

- As the writer:
  - Another reader's view: it shows your assumptions and blind spots
  - Fresh eyes on the structure: does the argument build?
  - A chance to add evidence and answer counter-arguments
  - A chance to fix style and tone for the audience
- As the reader:
  - You learn to weigh evidence and spot weak arguments
  - You see how other people solve the problems you have
  - Better writing, better discussion
- Think of the review you wrote on Moodle: which of these did it do?

Feedback as an ethical check

- Writing about people, cultures and data carries responsibility: a reader can catch bad framing, bias or harm you did not see
- Facts get checked, sources verified, sensitive topics handled with care
- Careful writing builds trust in scholarship as a whole

Roadmap

1. Feedback and Revision

2. Peer Review, Trust, and Metrics
   - Feedback Beyond Academia
   - Reviewing at the Association for Computational Linguistics

3. Problems: Bias & Inequality, (un)Reproducibility

What peer review is

- Experts evaluate a manuscript before it is published
- Its aims are to:
  - Check the **soundness** of methods and argument
  - Improve clarity, structure, and contribution
  - Build **trust** in published research
- Reviewers are gatekeepers and, at best, mentors

Types of peer review

- Three models, trading off transparency, fairness and accountability

Single-blind: Reviewers know the author's identity, but authors do not know the reviewers

- Common in science and social science
- Advantage: reviewers can assess expertise and potential conflicts
- Risk: unconscious bias (e.g., by gender, affiliation, language, or region)

Double-blind: Neither authors nor reviewers know each other's identity

- Reduces bias (see the WSDM experiment later)
- Often used in humanities and linguistics conferences
- Risk: less accountability, and reviewers can often guess the authors

Open review: Both sides know identities, and sometimes reviews are published

- Transparent, and reviewers get credit for their work
- Politer, but reviewers may hold back (see the BMJ trial later)
- Used in some open-science and data journals

Peer review: strengths and weaknesses

Strengths

- Catches errors before publication
- Provides expert feedback before publication
- A shared standard: “peer reviewed” means something
- Dialogue within a discipline

Weaknesses

- Can be slow, inconsistent, or biased
- Overburdens a small reviewer pool
- Reviewers rarely check replication
- Negative or null results often unpublished (the file-drawer effect)

The third reviewer

- Every academic has a story about the reviewer who sank their paper
  - Scientific Peer Review, ca. 1945: a *Downfall* parody — two good reviews, then the third reviewer wants more experiments
  - Shit My Reviewers Say: real review comments, collected
- The joke has since moved to **Reviewer 2**
- Peterson (2020) tested it on the reviews of one journal (*Political Behavior*):
  - Reviewer 2 is no more negative than the others
  - Reviewer 3 is the one most often far below the other reviewers
- The point for you: a harsh review is one reader's view — read it for what you can fix

Evolving models of peer review

- **Post-publication review:**
  - Articles or datasets are published first, then openly discussed
  - Examples: F1000Research, PubPeer
- **Community review:**
  - Large-scale collaborative commenting before publication (e.g., open preprints, GitHub issues, Open Review at LangSci Press)
  - Many eyes, in the open
- **Preprint feedback:**
  - Early sharing (e.g., arXiv, LingBuzz) gets informal feedback before submission
  - Faster, and it establishes priority (no fear of being scooped)
- In all three, review is a conversation that continues after publication

Feedback in industry: giving notes

- In film, publishing, games and advertising, feedback is called **notes** (Catmull & Wallace, 2014)
- Notes come from editors, producers, managers and clients, usually down a hierarchy
- The goal is the same as in academic review:
  - make the work better
  - make sure it does its job for its audience
- The constraints differ:
  - deadlines, budgets, brand guidelines, lawyers
  - the writer, the editor and the client may want different things

Video: Ed Catmull on the Pixar Braintrust (Stanford, 8 min)

How industry notes differ from academic peer review

- **Hierarchical**: the notes come from someone with authority over the project
- **Goal-driven**: the note serves the project, brand or product
- **Negotiated, up to a point**: creators argue back, but the final call may not be theirs
- **Iterative**: notes, revisions, more notes, approval
- **Success** means the work does what the project needs

Best practices from industry feedback culture

- **Clarity first**: say what the problem is before proposing a fix (the Pixar rule)
- **Respect the creator's intent**: improve the work without imposing your style
- **Prioritise**: not every note can be taken; choose the ones that serve the project
- **Negotiate**: creators can ask what a note means, and push back
- **Shared responsibility**: notes name problems; the creator chooses the fix

What students can learn from industry feedback

- You will get feedback from editors, managers, clients and colleagues
- What carries over:
  - receiving feedback without defensiveness
  - interpreting conflicting goals
  - deciding which suggestions to adopt, adapt, or reject
  - communicating clearly about revisions

Feedback cultures beyond academic peer review

- **Journalism**: editors and fact-checkers check claims against primary sources before publication (academic reviewers rarely check that a citation says what the paper claims)
- **Wikipedia**: articles pass through community review; Good and Featured Article criteria require verifiability and a neutral point of view
- **Code review**: peers review code for bugs, style and maintainability, in public and tied to version control (GitHub pull requests)

Peer review in Computational Linguistics

- Whole communities invest in improving peer review
- The Association for Computational Linguistics (ACL) publishes:
  - reviewer training guides (Boyd-Graber et al., 2023)
  - best-practice recommendations (ACL Rolling Review, 2025)
  - annual reports analysing what works and what fails (ACL Standing Committee on Peer Review, 2025)
- A large conference at scale — ACL 2025, for example (ACL 2025 Program Chairs, 2025):
  - 8,360 submissions
  - 1,699 accepted to the main conference (20.3%), 1,392 as Findings (16.7%)
  - 169 Senior Area Chairs, 1,937 Area Chairs, 11,720 reviewers!
  - Each paper reviewed by three reviewers, the authors respond, the reviewers update their reviews, the area chairs suggest which papers are accepted, the senior area chairs make the final decisions, …

Accept reviews only when appropriate (ACL guidelines)

- Before agreeing to review, check:
  - **Expertise**: Can you fairly assess the methods and claims?
  - **Conflicts of interest**: shared institution, collaborators, competitors
  - **Capacity**: do you have time to read it carefully?
- Accepting when you are unqualified or overloaded leads to unfair or rushed reviews
- Slightly undermined by the fact that if you submit a paper, someone from the author team must volunteer to review, and owes four reviews per submission (capped at 16 in a cycle), …
  - ACL 2025 accepted 20.3% to the main conference and a further 16.7% as Findings — four papers miss the main conference for every one that makes it, and nearly two thirds are turned down altogether
  - From October 2026 the ACL sustainable reviewing policy goes further:
    - each submission needs a qualified service contributor (at most two submissions each)
    - authors are capped at 20 submissions and 5 first-author submissions a cycle
    - no registered service, no review: the paper is desk rejected

Read carefully and generously (ACL guidelines)

- First read: understand the research question and contribution
- Second read: evaluate evidence, structure, and clarity
- Look for what the author is trying to do
- Avoid snap judgments based on style, formatting, or first impressions

Write constructive, actionable feedback (ACL guidelines)

- Every review should include both:
  - **Strengths**: what works
  - **Weaknesses**: what to fix
- Make criticisms specific:
  - point to a sentence or section
  - explain why it is unclear
  - suggest how to fix it
- Vague comments (“the paper is confusing”) do not help

Check evidence, ethics, and reproducibility (ACL guidelines)

- A reviewer is responsible for more than novelty
- Consider:
  - Are claims supported by evidence?
  - Are methods and data described clearly?
  - Are ethical implications acknowledged?
  - Can results be verified or reproduced?

Professionalism and respect (ACL guidelines)

- Use a constructive and collegial tone
- Avoid dismissive language:
  - “This is trivial”
  - “obviously wrong”
  - “the authors don’t understand”
- Remember: reviewers and authors are colleagues
- Treat the paper the way you would want yours to be treated

Reviewer power and responsibility (ACL COPR)

- Reviewers gain **access to unpublished work** before the community
- This creates responsibility:
  - Never share, quote or use ideas from the manuscript
  - Never delay a review to gain priority
  - Never punish papers that challenge your favourite theory
  - Declare conflicts of interest
- Rockwell (2005): a manuscript under review is confidential; do not share it, and do not use its data or ideas in your own work
- Peer review runs on trust

What this means for you (student reviewers)

- The same principles apply in the classroom:
  - read generously
  - comment specifically
  - focus on clarity, evidence, and logic
  - be respectful and supportive
- Practising good peer review improves your own writing:
  - you learn what makes an argument work
  - you see how readers interpret drafts
  - you build editorial judgment

Based on ACL reviewer training materials (ACL Rolling Review, 2025; ACL Standing Committee on Peer Review, 2025; Boyd-Graber et al., 2023).

ACL-inspired peer review checklist

- Am I qualified and conflict-free?
- Have I read the paper carefully (twice)?
- Did I identify strengths and weaknesses?
- Are my comments specific and actionable?
- Did I evaluate ethics, evidence, and reproducibility?
- Is my tone respectful and professional?
- Have I upheld confidentiality and integrity?

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   - Punktoza
   - Bias and Inequality in Peer Review
   - The Reproducibility Crisis
   - AI and Peer Review

Publish or Perish

- Academics are under pressure to publish
- This is a major part of how they are evaluated
- This should lead to more, higher quality publications
- But can lead to many more, lower quality publications
  - Or even fraud!
- And means we have to review more and more publications

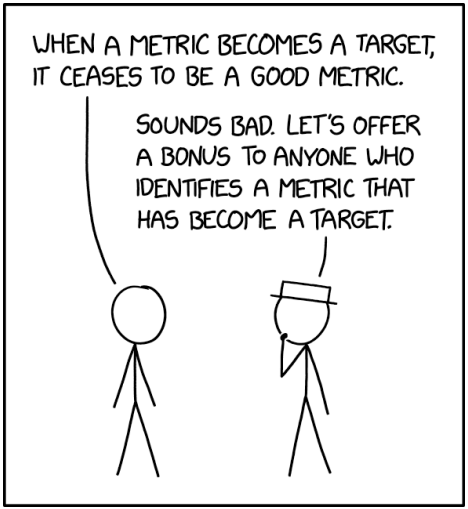
Metrics: why they were introduced

- Evaluators wanted standard numbers to compare:
  - journals (Impact Factor)
  - researchers (h-index)
  - institutions (citation counts, productivity)
- **Impact Factor (IF):** citations made in one year to what a journal published in the *two preceding* years, divided by the number of “citable items” (research articles and reviews) it published in those two years
  - Designed to help libraries choose which journals to buy
  - A standard criticism is the mismatch between the two halves: the numerator counts citations to *everything* (editorials, letters, news), the denominator counts only the “citable items”
- **h-index:** a researcher has index *h* if they have *h* papers each cited at least *h* times
  - Productivity and influence in one number
- Meant to inform expert judgement, not to replace it

Metrics: when measures become targets

- When hiring, promotion, or funding is tied to metrics:
  - Researchers optimise for what is counted, not what is meaningful
  - Citation counts become currency; novelty outranks replication or rigour
- **Goodhart's Law**, as Strathern (1997, p. 308) put it:
  - "When a measure becomes a target, it ceases to be a good measure"
  - IF and h-index stop measuring anything once people game them
- **Punktoza** ("points fever"): publishing to collect evaluation points (Kulczycki, 2017a, 2017b)
  - Encourages salami-slicing, low-risk incremental work, quantity over substance
  - A rational response to points-based national evaluation

Goodhart's Law xkcd 2899



Peer review is not neutral

- Peer review should judge the work alone, but social biases shape outcomes
- Studies find bias by gender, prestige and affiliation (next two frames)
- The result: the same people stay visible

Gender bias and underrepresentation

- A study of 145 journals (1.7 million authors and 740,000 referees) found no penalty for manuscripts by women (if anything, referees and editors treated them slightly more favourably), yet women remain under-represented among reviewers and editors (Squazzoni et al., 2021)
- An earlier study of 9,000 editors and 43,000 reviewers at one publisher found the same under-representation, and that reviewers tend to be matched with authors like themselves (Helmer et al., 2017; Murray et al., 2018)
- In grant-review contexts, women have also been shown to receive lower scores despite similar qualifications (Wennerås & Wold, 1997)

Affiliation and prestige bias

- Affiliation bias has been measured directly: WSDM 2017 ran a controlled experiment in which every paper got two *single-blind* reviewers and two *double-blind* ones (Tomkins et al., 2017)
  - The single-blind reviewers were markedly more likely to recommend acceptance for papers by famous authors (odds $\times 1.63$) and from top universities ($\times 1.58$) or top companies ($\times 2.10$)
  - Their double-blind colleagues, reading the same papers, showed no such advantage — so double-blinding takes it away
- So publication records amplify institutional inequality: the rich get richer

Open review — which is what you just did

- Your reviews went in the Moodle forum: **signed**, and readable by the whole class. That is **open review**, and some journals work this way (eLife, BMJ, F1000).
- **What blinding buys you**: double-blinding removes the famous-author advantage (Tomkins et al., 2017).
- **What signing costs you**: in a randomised trial at the *BMJ*, reviewers asked to be identified to the author were **12 points more likely to refuse** to review at all (35% v 23%) — but for those who did review, quality, recommendation and time taken were **no different** (van Rooyen et al., 1999).

So, from your own week:

- Did you soften anything because your name was on it?
- Did reading other people's reviews change how you wrote yours?
- Would you have said more, or less, anonymously?

Why these biases happen, and what we can do

- Why these biases happen
  - Reviewers and editors lean on status cues: institution, reputation, familiarity
  - Reviewing is human work, and unconscious bias (gender, prestige, nationality, career stage) is common
  - It is structural too: who is asked to review, whose papers get seen, who gets accepted
- What this means for us
  - As reviewers: judge the content, and notice when you are judging the name
  - As authors: a rejection is not only about quality
  - As a community: diverse reviewers and editors, anonymised review where it fits, transparent editorial decisions

What is reproducibility / replicability?

- **Reproducibility:** re-analyse the same data/code to get the same result
- **Replicability:** collect new data under same conditions and test whether the effect holds
- Without either, a published finding is provisional (Goodman et al., 2016)

The crisis in numbers

- In a 2015 project replicating 100 psychology studies, only about 36% of results were successfully replicated; effect sizes dropped by about 50% (Open Science Collaboration, 2015)
- A 2016 *Nature* survey of 1,576 researchers, across all the sciences, found over 70% had tried and failed to reproduce another scientist's results, and 52% judged that there is a significant crisis (Baker, 2016)
- Cancer biology is no better: an eight-year project repeated 50 experiments from 23 high-profile papers, and only 46% of the replications (51/112) succeeded, with effect sizes a median 85% smaller (Errington et al., 2021)
- If findings routinely fail to replicate, the literature misleads everyone who builds on it

Video: Veritasium: "Is Most Published Research Wrong?" (12 min)

Causes: why reproducibility fails

- Low statistical power (too few subjects): effects that come and go (Ioannidis, 2005)
- Publication bias: positive, novel results get published; null results stay in the drawer
- Missing code, data and method details: honest researchers make replication impossible by not sharing
- Incentives: rewards go to novelty, impact and speed, none to replication

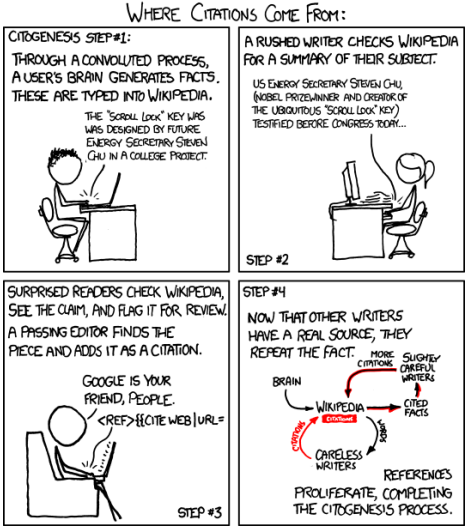
FAIR, CARE and good writing help

- FAIR: metadata, shared code and data make re-analysis possible
- CARE: honest reporting of context and limitations
- Good writing: a reader can tell exactly what was done
- Together: more reliable science

Citation as trust

- A citation says: I read this and trust it enough to build on it
- Citations add up to networks of credibility
- Not all citations are equal:
  - Some endorse, others critique or merely mention
  - Citation cascades pass on claims no one checked (next slide)
- Cite what you have read, and say what you take from it

Citogenesis xkcd 978



AI in peer review: a known problem

- Large language models (LLMs) are used in academic writing, and in peer review
- A 2024 study estimated that 6.5–17% of peer reviews at major AI conferences may have been substantially written or modified by LLMs (Liang et al., 2024)
- The tell: *commendable*, *meticulous* and *intricate* rose 10–35 times in frequency
- Major publishers all restrict AI in peer review, but not in the same way:
  - **All of them:** never upload the manuscript, or any part of it, to a public AI tool: that alone breaks confidentiality
  - **Taylor & Francis:** AI must not generate the review; it may only improve the language of a review you wrote yourself
  - **Sage:** AI may help write and format your review, but not generate it from the manuscript; you remain responsible for its content
  - **Springer Nature:** AI may support a reviewer but never replace their judgement; use only secure or institutionally approved tools, and be transparent about it

Why AI-generated reviews are problematic

- **Confidentiality breach**: uploading a manuscript to an LLM shares it with a third party
- **Accountability**: reviewers answer for accuracy and judgement, and an LLM cannot
- **Bias and hallucination**: LLMs repeat biases and produce plausible, wrong critiques
- **Superficiality**: generated reviews are generic and miss the issues an expert would see
- **Trust erosion**: an automated review is no longer an expert's judgement

Typical guidance on AI in peer review

- Most journals and conferences now require:
  - Disclosure of any AI assistance in writing or reviewing
  - Human accountability for all submitted content
  - No uploading of confidential manuscripts to AI tools
- Acceptable uses (with disclosure):
  - Grammar and spell-checking
  - Improving clarity of your own writing
- Unacceptable uses:
  - Generating review text
  - Summarising or analysing manuscripts you are reviewing
  - Submitting AI-generated content as your own expert judgment

Acknowledgements

- I have had over 200 papers reviewed (thanks reviewers!)
- I have peer reviewed over 2,000 conference papers, 100 journal papers, 10 PhDs, 2 Habilitations, and many grants
- **GPT-4o** in ChatGPT (OpenAI, 2025) was used to format the references, and generate a first draft of the slides
- **Claude Opus 5** (Anthropic, 2026) was used to check the slides: to resolve every DOI against Crossref, fetch every URL, and compare each quotation with its original. The prompt was of the form

  ```
  Verify every claim, DOI, URL and quotation against a real source. Where you cannot
  verify something, say so rather than guessing --- never invent a reference.
  ```

  Each correction names the source it was checked against. Responsibility for what remains is mine.

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